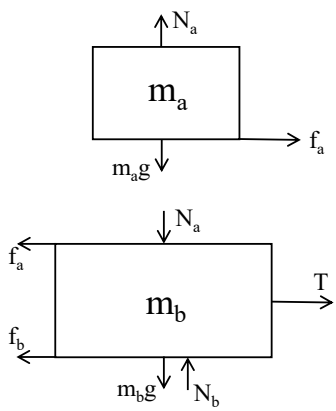
Solution to CAP 2023

1 Long Question

Q1

Consider the following free body diagrams for the boxes m_a and m_b:



Horizontally,

$$f_a = m_a a_a \tag{1}$$
$$T - f_a - f_b = m_b a_b \tag{2}$$

where f_a and f_b are the frictional forces, a_a and a_b are the accelerations of the boxes m_a and m_b.

(1 point for relating horizontal components)

Vertically,

$$N_a = m_a g \tag{3}$$
$$N_b - N_a - m_b g = 0 \tag{4}$$

(1 point for relating vertical components)

(a)

When the two boxes are static, $a_a = a_b = 0$ and $f_b \leq \mu_b N_b$.

By (2), $T = f_b \leq \mu_b N_b$

(1 point for conditions for the boxes to be static)

Using (4), $T \leq \mu_b(N_a + m_b g) = \mu_b(m_a + m_b)g$

Therefore, $T \leq \mu_b(m_a + m_b)g$.

(1 point for the correct inequality for the range of T)

(b)

In this case, $f_a \leq \mu_a N_a$, $f_b = \mu_b N_b$ and $a_a = a_b$.

(1 point for conditions that have to be satisfied)

By (1), $a_a = \frac{f_a}{m_a}$

By (2), $a_b = \frac{T - f_a - f_b}{m_b}$

Putting these into $a_a = a_b$, we have

$$\begin{aligned}
\frac{f_a}{m_a} &= \frac{T - f_a - f_b}{m_b}. \\
\frac{T - f_b}{m_b} &= f_a(\frac{1}{m_a} + \frac{1}{m_b}) \\
(T - f_b)\frac{m_a}{m_a + m_b} &= f_a \leq \mu_a N_a \\
(T - \mu_b(m_a g + m_b g))\frac{m_a}{m_a + m_b} &\leq \mu_a m_a g \\
T &\leq \mu_a g(m_a + m_b) + \mu_b(m_a + m_b)g = (\mu_a + \mu_b)(m_a + m_b)g
\end{aligned}$$

(1 point for algebra work of trying to use the vertical and horizontal relations to work out the range of T)

But note, by part (a), T has to be $\geq \mu_b(m_a + m_b)g$ such that the boxes are in motion. So we have

$$\mu_b(m_a + m_b)g \leq T \leq (\mu_a + \mu_b)(m_a + m_b)g$$

(1 point each for the correct lower and upper bound of T (2 points in total))

(c)

When there is relative motion between the two boxes, we have

$$\begin{cases} f_a &= \mu_a N_a = \mu_a m_a g \\ f_b &= \mu_b N_b = \mu_b(N_a + m_b g) = \mu_b(m_a + m_b)g \end{cases}$$

and $a_b > a_a$.

(1 point for the conditions that have to be satisfied)

By (1) and (2),

$$\begin{cases} a_a = \frac{f_a}{m_a} = \frac{\mu_a m_a g}{m_a} = \mu_a g \\ a_b = \frac{T - f_a - f_b}{m_b} = \frac{T - \mu_a m_a g - \mu_b(m_a + m_b)g}{m_b} \end{cases}$$

which gives $\frac{T-\mu_a m_a g-\mu_b(m_a+m_b)g}{} > \mu_a g$.

Therefore, $T > (\mu_a + \mu_b)(m_a + m_b)g$

(1 point for correct inequality for the range of T)

Q2

(1 point for recognizing this is a conservation of energy and momentum question)
Conservation of energy:

$$\frac{hc}{\lambda} - \frac{hc}{\lambda'} = mc^2 - m_o c^2 \tag{1}$$

Conservation of momentum:
In the x direction:

$$\frac{h}{\lambda} = \frac{h}{\lambda'} cos\theta + mvcos\phi \tag{2}$$

In the y direction:

$$\frac{h}{\lambda'} sin\theta = mvsin\phi \tag{3}$$

(1 point each for correctly getting equations 1, 2, 3 (3 marks total))
Square equation [2] and [3] (1 mark)

$$m^2 v^2 cos^2\phi = h^2(\frac{1}{\lambda^2} - \frac{2cos\theta}{\lambda\lambda'} + \frac{cos^2\theta}{\lambda^2}) \tag{4}$$

$$m^2 v^3 sin^2\phi = h^2(\frac{sin^2\theta}{\lambda^2}) \tag{5}$$

Add together to eliminate ϕ: (2 marks)

$$m^2 v^2 (sin^2\phi + cos^2\phi) = h^2(\frac{1}{\lambda^2} - \frac{2cos\theta}{\lambda\lambda'} + \frac{1}{\lambda^2}) \tag{6}$$

Since $m^2 v^2 = c^2(m^2 - m_o^2)$ we get:

$$c^2(m^2 - m_o^2) = h^2(\frac{1}{\lambda^2} - \frac{2cos\theta}{\lambda\lambda'} + \frac{1}{\lambda^2}) \tag{7}$$

(1 mark for getting rid of v)
Solve [1] for m and square:

$$m^2 = m_o^2 + \frac{2m_o h}{c}(\frac{1}{\lambda} - \frac{1}{\lambda'}) + \frac{h^2}{c^2}(\frac{1}{\lambda} - \frac{1}{\lambda'})^2 \tag{8}$$

(1 mark for last substitution and solving)
Sub [5] into [4]:

$$2m_o hc(\frac{1}{\lambda} - \frac{1}{\lambda'}) + h^2(\frac{1}{\lambda^2} - \frac{2}{\lambda\lambda'} + \frac{1}{\lambda^2}) = h^2(\frac{1}{\lambda^2} - \frac{2cos\theta}{\lambda\lambda'} + \frac{1}{\lambda^2}) \tag{9}$$

Cancel terms and multiply both sides by $\frac{\lambda\lambda'}{h}$

$$m_o c(\lambda' - \lambda) - h = -hcos\theta \tag{10}$$

$$\lambda' - \lambda = \frac{h}{m_o c}(1 - cos\theta) \tag{11}$$

(1 mark for correct final answer)

Q3

When you look at the first drawing of the sensor you realise that the shortest measurable distance will be when the returning beam hits the top most pixel of the linear sensor; similarly the longest distance is measured when the returning beam hits the lowest pixel. For the longest measurable distance we can look at the similar triangles in front and in the back of the slit and notice that the the tangents of the same angle give us:

$$\frac{0.015}{2} = \frac{5}{d}$$
$$\therefore d_{max} = 667mm$$

Similarly, for a shortest distance we have:

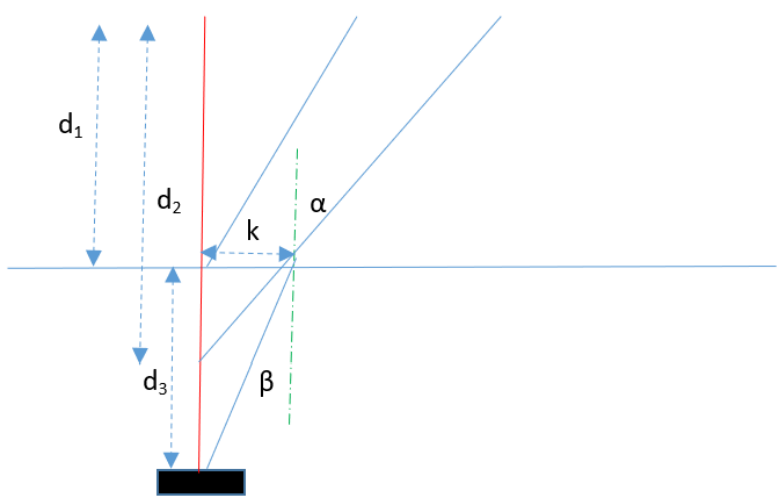
$$\frac{0.015 \times 600}{2} = \frac{5}{d}$$
$$\therefore d_{min} = 1.1mm$$

Generally knowing that pixel n counting from the bottom was activated we get the distance

$$d = \frac{10}{n} \times 0.015\ mm$$

If we measure the distance to the object in the water we get two pixels activated: n_1 the one corresponding to the distance d_1 to the water and n_2 corresponding to the apparent distance d_2 to the object under water.

From Snells law we have:



$$\frac{\sin\alpha}{\sin\beta} = n$$

where n is the refraction index of water From the picture:

$$\tan\alpha = k/(d_2 - d_1)$$
$$\tan\beta = k/d_3$$

Assuming that all the distances are much bigger than the sensor we can approximate both sine and tangent functions of the angles by the angle values:

$$\frac{d_3}{d_2 - d_1} = n$$
$$d_3 = n(d_2 - d_1)$$

distance d to the object under water is:

$$d = d_1 + d_3 = d_1 + n(d_2 - d_1)$$

Marking scheme Q3

Step	Points
Being able to use triangular principle to calculate distances	1
Identifying that the minimum distance is found when the light hits the highest pixel	1
Correctly finding the minimum distance	1
Identifying that the maximum distance is found when the light hits the lowest pixel	1
Correctly finding the maximum distance	1
Reasoning that the underwater distance is to be 'corrected' due to refraction	1
Correctly writing Snell's law	1
Setting up an equation for the modified distance	1
Correct manipulation and final result	2

Short Question

1.

(D)

The tension is 30N all along the wire because of Newton's third law. Since the cross-section varies, the stress is different at different cross-sections.

2.

(B)

Key concept: the change in the mechanical energy of a body is equal to the work done by the non-conservative forces acting on the body.
Height of ramp $= 3.6\sin(35°) = 2.1m$
$W_{friction} = [\frac{1}{2}mv_i^2 + mgh_i] - [\frac{1}{2}mv_f^2 + mgh_f] = [\frac{1}{2}(1)(2)^2 + (1)(9.8)(2.1)] - [\frac{1}{2}(1)(1.06^2)] =$ 21.7 Nm

3.

(A)

The banana and monkey A continue to travel tangential to the point where he releases the vine. Therefore, to monkey A the banana appears stationary immediately after he releases the vine and drops the banana, and to monkey B it follows a path tangential to the rotational motion from swinging on the vine.

4.

(E)

Conservation of energy:
Since the ball starts at the same height and moves at the same initial speed, we can use conservation of energy to find the speed of the shell when it hits the ground.
$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + mgh$
$\therefore v = \sqrt{v_0^2 + 2gh}$ in all cases.

Alternate solution:
$v_A = v_B = v_C = v_0$
Throw upwards: $v_{A,fin} = \sqrt{v_0^2 + 2gh}$
Throw downwards: $v_{C,fin} = \sqrt{(-v_0)^2 + 2gh} = \sqrt{v_0^2 + 2gh}$
Throw horizontally:
Initial vertical velocity is 0, constant horizontal velocity is $v_{Bx} = v_B = v_0$
Final vertical: $v_{By} = -\sqrt{2gh}$

Total: $v_{B,fin} = \sqrt{v_{Bx}^2 + v_{By}^2} = \sqrt{v_0^2 + 2gh}$

5.

(E)
The x component of the velocity after collision is $v_x = -v\cos\alpha$ since the collision is elastic. To find the y component, we have to consider whether the ball is going upwards or downwards at the time of the collision depending on the value of t. If the ball is moving downwards just before the collision, the velocity in the box's frame (which is the center of mass frame since $M \gg m$) is $v^* = -|v_y| - u$, where $v_y = v\sin\alpha - gt$. It doesn't change after the collision. So, the velocity after the collision

$$v' = (-v\cos\alpha, |v\sin\theta - gt|)$$

Similarly, if the ball is moving upwards just before the collision, the velocity after the collision,

$$v' = (-v\cos\alpha, -|v\sin\theta - gt|)$$

6.

(A)

Since the sphere starts from an equilibrium position, the effect of gravity does not have to be considered in the subsequent motion.

Speed at equilibrium position $= v_{max} = 0.5$.

$$\begin{aligned} a_{max} &= v_{max}\sqrt{\frac{k}{m}} \\ &= 0.5\sqrt{\frac{3.2}{0.2}} \\ &= 2 \text{ ms}^{-1} \end{aligned}$$

7.

(C)

The speeds at point B and C are given by

$$\begin{aligned} v_B^2 &= v_A^2 + 2as \\ v_C^2 &= v_B^2 + 2as \end{aligned}$$

where a is the acceleration and s is the distance travelled between the neighboring points.
Eliminating $2as$ from the system of equations above, we can get

$$v_B = \sqrt{\frac{v_A^2 + v_C^2}{2}}$$

8.

(D)

The charges on X and Y can both be positive or negative, so (A) may not be correct.

Irrespective of the charges on the spheres, the forces on them must be of equal magnitude (and opposite directions), because of Newton's third law. So the only thing that can make one sphere lower than the other is their masses.

9.

(E)

The field at the center is given by

$$B = \frac{\mu_0 I}{2r}$$

since $I = \frac{E}{R}$ where R is the resistance of the wire, one sees that B depends on both the e.m.f and the radius r of the wire with a fixed resistance.

10.

(D)

Because of symmetry the field at directions A and B is zero. Since the semi-circle is negatively charged the net field is along D direction.

11.

(A)

The initial kinetic energy of the positron is converted to the electrical potential energy at the distance of minimum approach d_{min}:

$$\begin{aligned} \frac{1}{2} m_e v_0^2 &= \frac{e^2}{4\pi\epsilon_0 d_{min}} \\ d_{min} &= \frac{e^2}{2\pi\epsilon_0 m_e v_0^2} \end{aligned}$$

12.

(D)
The magnetic field inside the solenoid is parallel to it's axis. Since

$$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}\,,$$

there is no force on the electron and it will not deflect at all.

13.

(B)
The formula is

$$r = \frac{mv}{qB}$$

So decreasing q will increase r.

14.

(B)

The total flux through the cube is $24Q/\epsilon_0$. If the charge had been at the center of the cube, the flux through the bottom surface would have been one-sixth of the total flux i.e $4Q/\epsilon_0$. Since, the given flux is less than that, the charge must be placed slightly away from the bottom surface i.e towards the top surface, slightly off the center.

15.

(A)

Suppose the new equivalent resistance is R'_{AB}. Observe that $R_{AB} = R'_{AB}||1\Omega$. When a resistance is added in parallel, the equivalent resistance decreases. So $R_{AB} < R_{AB'}$ i.e $R'_{AB} > R_{AB}$.

16.

(C)

The diode is on as long as $V_{in} < -2V$. When it is on, the output voltage V_{out} is just the DC battery's voltage i.e $-2V$. When the diode is off, V_{out} is the AC source voltage.

17.

(B)

For adiabatic process $\Delta Q = 0$, consequently $\Delta U = \Delta W$. Since the gas expands $\Delta W > 0$. So we must have $\Delta U < 0$. Only option (B) satisfies all these criteria.

18

(E)

$$PV = nRT$$

T decreases so P decreases inside the cup.

19.

(C)

Use $PV = nRT$ to obtain

$$TV^2 = const$$

So, $T_2/T_1 = (V_1/V_2)^2 = 1/4$ when $V_2 = 2V_1$.

20.

(B)

$$\Delta Q = \Delta U + \Delta W$$

When the volume is constant, no work is done by the gas. So the amount of heat needed is less.

21.

(B)

Each atom at the corner of the unit cell is shared by eight other unit cells. So, the mass of the unit cell is

$$\left(\frac{1}{8}m\right) \times 8 + m = 2m$$

where m is the mass of the Na atom. m can be found from the atomic mass $M_A = 23g/mol$ via

$$m = M_A/N_A$$

where N_A is the Avogadro's number. So, density

$$\rho = \frac{2M_A}{a^3 N_A} = 0.94\ g/cm^3.$$

22.

(B)

The refractive index (n) is inversely proportional to wavelength (λ), or proportional to frequency. Therefore, in the Snell's law, for the same incident angle from a less dense medium (air) into the glass, the component with higher frequency has a larger n and thus a smaller refracted angle θ. So only b) and d) are possible.

When the ray comes out, using the Snell's law, the angles in air should be the same as that before the ray go into the glass. So only b) is left.

23.

(C)

Red light has lower frequency (i.e. lower energy) than green light. Therefore if the photocell can emit electrons when illuminated by green light, it also emits electrons when it is shined by red light. However, ultraviolet, X-rays and blue light have higher frequency than green light, so we are uncertain if the photocell emits electrons when it is illuminated by them.

24.

(E)

The intensity of the bright and dark spots are

$$\begin{aligned} I_{max} &= I_1 + I_2 + 2\sqrt{I_1 I_2} \\ I_{min} &= I_1 + I_2 - 2\sqrt{I_1 I_2} \end{aligned} \tag{12}$$

Initially, with $I_1 = I_2 = I$ we have

$$I_{max} = 4I, \quad I_{min} = 0$$

. Now, if $I_1 = I$ and $I_2 = I/2$ we get

$$I_{max} = \frac{1}{2}(3 + 2\sqrt{2})I, \quad I_{min} = \frac{1}{2}(3 - 2\sqrt{2})I$$

So, the bright fringe has gone dimmer and the dark fringe becomes brighter.

25.

(C)

Recall the standard formula of the spacing in air

$$\Delta x = \frac{\lambda R}{d}.$$

When the system is submerged in water only the wavelength changes as $\lambda \to \frac{\lambda}{n_w}$. So,

$$\Delta x' = \frac{\lambda R}{d n_w}.$$

Question 1	a	b	c	d	e	f
Question 2	a	b	c	d	e	f
Question 3	a	b	c	d	e	f
Question 4	a	b	c	d	e	f
Question 5	a	b	c	d	e	f
Question 6	a	b	c	d	e	f
Question 7	a	b	c	d	e	f
Question 8	a	b	c	d	e	f
Question 9	a	b	c	d	e	f
Question 10	a	b	c	d	e	f
Question 11	a	b	c	d	e	f
Question 12	a	b	c	d	e	f
Question 13	a	b	c	d	e	f
Question 14	a	b	c	d	e	f
Question 15	a	b	c	d	e	f
Question 16	a	b	c	d	e	f
Question 17	a	b	c	d	e	f
Question 18	a	b	c	d	e	f
Question 19	a	b	c	d	e	f
Question 20	a	b	c	d	e	f
Question 21	a	b	c	d	e	f
Question 22	a	b	c	d	e	f
Question 23	a	b	c	d	e	f
Question 24	a	b	c	d	e	f
Question 25	a	b	c	d	e	f